SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Department of Information Technology

CSA51 – Cryptography and Network Security

ASSIGNMENT 1 – CO1

Design, Analyse and Validate Classical Encryption Techniques for Securing Sensitive Information

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
RAKSHITH R.R	192324158	B-TECH-AIDS	4th year/8th	Dhamani.R	Individual	1 Week	50	10.10.26	11.10.26

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
1	Cipher Design and Selection	5	5	Select and describe three suitable classical ciphers (Caesar, monoalphabetic substitution, and Vigenère or Hill); justify their use and limitations for the given sensitive message.
2	Encryption and Decryption	5	5	Show correct key selection, encryption steps, ciphertext, decryption steps and recovery of the original plaintext.
3	Key Analysis and Mathematical Basis	5	5	Explain key space, modular arithmetic or alphabet mapping used by each selected cipher; calculations must be traceable.
4	Security and Cryptanalysis	5	5	Analyse vulnerabilities such as brute-force search, frequency analysis, repeated-key patterns and known-plaintext attacks where relevant.
5	Comparison of Cipher Strength	4	4	Compare the selected methods using key space, implementation effort, likely attacks and suitability for sensitive information.
6	Algorithm / Pseudocode Formulation	4	4	Provide clear encryption and decryption pseudocode or flowcharts, including input validation and treatment of spaces/punctuation.
7	Worked Example and Verification	4	4	Use a sample message; verify that decrypt(encrypt(plaintext)) reproduces the original message under the stated rules.
8	Implementation and Testing	4	4	Implement at least two ciphers in a programming language; include source code, test cases, expected output and actual output.
9	Experimental Evaluation	3	3	Test multiple messages and key choices; present organised tables of ciphertext, recovered plaintext, execution time or operation counts.
10	Theory-to-Result Discussion	3	3	Explain how observed results match the cipher design and why classical ciphers are not suitable for modern high-security use.
11	Recommend and Defend a Design	4	4	Recommend a method for the assignment scenario and defend the decision using correctness, usability and security limitations.
12A	Re-evaluate: Impact Analysis	2	2	Discuss the impact of message length, key choice and repeated patterns on confidentiality.
12B	Re-evaluate: Attack Evaluation	2	2	Evaluate likely attacks against the selected ciphers and identify information an attacker could exploit.
12C	Re-evaluate: Adaptation or Redesign	2	2	Propose a safer modern replacement or layered approach, explaining why classical ciphers are only educational examples.
12D	Re-evaluate: Evidence-Based Defence	2	2	Defend conclusions using worked examples, test results, key-space reasoning and a comparison table.
	TOTAL	50	50	Marks to be awarded by the faculty.

OVERALL REMARKS

Good. Assignment well presented.

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Faculty feedback:

Student Signature	Faculty Signature	Program Director

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present ✗ = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
1	Cipher selection, design rationale and assumptions	☑	Three ciphers described and justified
2	Encryption and decryption working shown	☑	Steps, key and plaintext/ciphertext included
3	Mathematical basis and key-space explanation	☑	Calculations are traceable
4	Security analysis and likely attacks	☑	Relevant weaknesses identified
5	Cipher comparison table	☑	Criteria applied consistently
6	Pseudocode or flowchart	☑	Encryption and decryption steps included
7	Worked example and correctness verification	☑	Recovered plaintext matches original
8	Source code and reproducible tests	☑	Program, test inputs and outputs included
9	Experimental results in tables	☑	Multiple inputs or keys evaluated
10	Theory and experimental results compared	☑	Observations interpreted
11	Recommendation supported by analysis	☑	Trade-offs and limitations stated
12A–D	Impact, attack evaluation, redesign and evidence-based defence	☑	All four parts addressed

IMPLEMENTATION EVIDENCE CHAIN (Tasks 8–9)

Evidence Item	Status	Observation
Source code	☑	Readable source code for each implemented cipher; language and execution instructions stated
Test inputs	☑	Plaintexts, keys and edge cases recorded
Expected behaviour	☑	Expected ciphertext and recovered plaintext documented
Actual output	☑	Program output or screenshots included
Intermediate evidence	☑	Key mapping, modular steps or intermediate transformations shown
Boundary / failure tests	☑	Empty input, spaces, punctuation, invalid keys and repeated characters considered
Interpretation	☑	Results connected to cipher properties and security limitations
Code, data and report consistent	☑	Tables and claims match the submitted program output

SUBMISSION CHECKLIST (not separately marked)

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Requirement	Status
All assignment tasks including 12A–12D answered	☑ Yes ☐ Partly ☐ No
Encryption and decryption shown for each selected cipher	☑ Yes ☐ Partly ☐ No
Keys and assumptions clearly stated	☑ Yes ☐ Partly ☐ No
At least two ciphers implemented and tested	☑ Yes ☐ Partly ☐ No
Same plaintext conventions used consistently	☑ Yes ☐ Partly ☐ No
Source code and reproducible test procedure provided	☑ Yes ☐ Partly ☐ No
Tables / figures / program outputs included	☑ Yes ☐ Partly ☐ No
Cross-references and numbering correct	☑ Yes ☐ Partly ☐ No
No unfilled template placeholders in the student report	☑ Yes ☐ Partly ☐ No
Viva recommended where implementation or calculations need clarification	☑ Yes ☐ No